LA HW

Included exercises: 1, 3, 7, 9, 13, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 41, 47, 49

Exercises 1–3: Product Dimensions

Decide whether each matrix product AB is defined. If so, find its size.

Original Exercise 1 A is a 2×3 matrix, and B^T is a 2×3 matrix.

Original Exercise 3 A^T is a 3×3 matrix, and B is a 2×3 matrix.

Exercises 5–20: Matrix Products

Compute each expression, or give a reason why it is not defined, using

$$A = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 7 & 4 \\ 1 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 3 & 8 & 1 \\ 2 & 0 & 4 \end{bmatrix},$$

$$\mathbf{x} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \qquad \mathbf{y} = \begin{bmatrix} 1 \\ 3 \\ -5 \end{bmatrix}, \qquad \mathbf{z} = \begin{bmatrix} 7 & -1 \end{bmatrix}.$$

Original Exercise 7 Compute $\mathbf{xz}$.

Original Exercise 9 Compute $AC\mathbf{x}$.

Original Exercise 13 Compute BC.

Original Exercise 19 Compute C^2.

Exercises 21–24: Verify Matrix Identities

Use the matrices A, B, C, and $\mathbf{z}$ given above.

Original Exercise 21 Verify that $I_2C = CI_3 = C$.

Original Exercise 23 Verify that $(AB)^T = B^TA^T$.

Exercises 25–32: Entries and Columns

Compute only the requested entry or column, without computing the entire product, using

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ -3 & -2 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 0 \\ 4 & 1 \\ 3 & -2 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 1 & -1 \\ 4 & 3 & -2 \end{bmatrix}.$$

Original Exercise 25 Find the $(3, 2)$-entry of AB.

Original Exercise 27 Find the $(2, 3)$-entry of CA.

Original Exercise 29 Find column 2 of AB.

Original Exercise 31 Find column 1 of CA.

Exercises 33–50: True or False

Determine whether each statement is true or false.

Original Exercise 33 The product of two $m \times n$ matrices is defined.

Original Exercise 35 For any matrices A and B, if AB and BA are both defined, then $AB = BA$.

Original Exercise 37 If A and B are matrices, then both AB and BA are defined if and only if A and B are square matrices.

Original Exercise 41 If AB is defined, then the (i, j)-entry of AB equals $a_{ij}b_{ij}$.

Original Exercise 47 If AB is defined and AB is a zero matrix, then either A or B is a zero matrix.

Original Exercise 49 The product of two diagonal matrices is a diagonal matrix.